

INHA UNIVERSITY TASHKENT

DEPARTMENT OF CSE & ICE

Fall 2023 – Application Programming in Java

Lab Assignment # 2

Topic covered Arithmetic Operator and I/O in Java

Last Date of Submission: 27/09/2023

INSTRUCTIONS:

- Program should be well commented.
- Program should be easy to use.
- All Lab assignments are to be completed by an individual student on his/her computer system and not in groups

Problems :

P.1 Class Grader

The certain instructor assigns letter grade for his course based on the following table:

Score	Grade
≥ 90	A+
≥ 85	A
≥ 80	B+
≥ 75	B
≥ 65	C+
≥ 60	C
≥ 55	D+
≥ 50	D
< 50	F



figure (b)

Write a class, Grader, which has an instance variable, **score**, an appropriate **constructor** and **appropriate methods**.

Method **letterGrade()** that returns the letter grade as a String. Now write a demo class to test the Grader class by reading a score from the user, using it to create a Grader object after validating that the value is not negative and is not greater than 100. Finally, call the letterGrade() method to get and print the grade. See figure (b) for sample run.

P.2 Huge Integer Class

Create a class `HugeInteger` which uses a 40-element array of digits to store integers as large as 40 digits each.

Provide methods **`parse`**, **`toString`**, **`add`** and **`subtract`**. Method `parse` should receive a `String`, extract each digit using method `charAt` and place the integer equivalent of each digit into the integer array. For comparing `HugeInteger` objects, provide the following methods: **`isEqualTo`**, **`isNotEqualTo`**, **`isGreaterThan`**, **`isLessThan`**, **`isGreaterThanOrEqualTo`** and **`isLessThanOrEqualTo`**. Each of these is a predicate method that returns true if the relationship holds between the two `HugeInteger` objects and returns false if the relationship does not hold. Provide a predicate method `isZero`. If you feel ambitious, also provide methods `multiply`, `divide` and `remainder`.

[Note: Primitive boolean values can be output as the word “true” or the word “false” with format specifier `%b`.]

P.3 Date Class

Create class `Date` with the following capabilities:

a) Output the date in multiple formats, such as

`MM/DD/YYYY`

`June 14, 1992`

`DDD YYYY`

b) Use overloaded constructors to create `Date` objects initialized with dates of the formats in part (a).

In the first case the constructor should receive three integer values. In the second case it should receive a `String` and two integer values. In the third case it should receive two integer values, the first of which represents the day number in the year.

[Hint: To convert the `String` representation of the month to a numeric value, compare `Strings` using the `equals` method. For example, if `s1` and `s2` are `Strings`, the method call `s1.equals(s2)` returns true if the `Strings` are identical and otherwise returns false.]
